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Erratum: Heavy quarkonium dynamics at next-to-leading order in the binding energy over temperature

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The end of section 2.4 starting from "The decay width of the state $\rho_u^l(r)$ is given by..." until the end of the section should be replaced by

> The decay width of the state $\rho_u^l(r)$ is given by the trace of the anti-commutator term of the Lindblad equation

$$\Gamma\left[\rho_u^l(r)\right] \equiv \sum_{n=0}^{1} \mathrm{Tr}\left[C^{n\,\dagger} C^n \rho_u^l(r)\right], \tag{2.41}$$

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where on the left $\Gamma[\rho_u^l(r)]$ represents a functional returning the width and on the right the trace is over color, angular momentum, and position. The partial widths with respect to each of the color channels are given by

$$\Gamma_{u\to v}\rho^l(r) = \sum_{n=0}^{1} \mathrm{Tr}_c \left[C^{n\,\dagger} |v\rangle\langle v| C^n \rho_u^l(r) \right], \tag{2.42}$$

where the trace is only over color, and $|v\rangle\langle v|$ is a state of color v. Computing the operators explicitly, we find

$$\Gamma_{s\to o} = \kappa \left\{ r^2 \left(1 + \frac{\Delta V_{so}}{4T}\right)^2 - \frac{3}{2MT} - \frac{\Delta V_{so}}{4MT^2} + \frac{\mathcal{D}^2}{4M^2T^2} \right\}, \tag{2.43}$$

$$\Gamma_{o\to s} = \frac{\kappa}{N_c^2-1} \left\{ r^2 \left(1 + \frac{\Delta V_{os}}{4T}\right)^2 - \frac{3}{2MT} - \frac{\Delta V_{os}}{4MT^2} + \frac{\mathcal{D}^2}{4M^2T^2} \right\}, \tag{2.44}$$

$$\Gamma_{o\to o} = \frac{\kappa(N_c^2-4)}{2(N_c^2-1)} \left\{ r^2 - \frac{3}{2MT} + \frac{\mathcal{D}^2}{4M^2T^2} \right\}, \tag{2.45}$$

where

$$\mathcal{D}^2 = -\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} \right) + \frac{l(l+1)}{r^2}\,. \tag{2.46}$$

The whole section 3.2 should be replaced by

When expressed as operators acting on the reduced wave function, the singlet-to-octet, octet-to-singlet and octet-to-octet width operators take the form

$$\overline{\Gamma}^{\uparrow}_{s\to o} = \hat{\kappa}T^3 \frac{l+1}{2l+1} \left[\left(r - \frac{N_c\alpha_{\rm s}}{8T} \right)^2 - \frac{2l+3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} + \frac{l+1}{2MT} \left(\frac{N_c\alpha_{\rm s}}{4T} \right) \frac{1}{r} \right], \tag{3.7}$$

$$\overline{\Gamma}^{\downarrow}_{s\to o} = \hat{\kappa}T^3 \frac{l}{2l+1} \left[\left(r - \frac{N_c\alpha_{\rm s}}{8T} \right)^2 + \frac{2l-1}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} - \frac{l}{2MT} \left(\frac{N_c\alpha_{\rm s}}{4T} \right) \frac{1}{r} \right], \tag{3.8}$$

$$\overline{\Gamma}^{\uparrow}_{o\to s} = \frac{\hat{\kappa}T^3}{N_c^2-1} \frac{l+1}{2l+1} \left[\left(r + \frac{N_c\alpha_{\rm s}}{8T} \right)^2 - \frac{2l+3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} - \frac{l+1}{2MT} \left(\frac{N_c\alpha_{\rm s}}{4T} \right) \frac{1}{r} \right], \tag{3.9}$$

$$\overline{\Gamma}^{\downarrow}_{o\to s} = \frac{\hat{\kappa}T^3}{N_c^2-1} \frac{l}{2l+1} \left[\left(r + \frac{N_c\alpha_{\rm s}}{8T} \right)^2 + \frac{2l-1}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} + \frac{l}{2MT} \left(\frac{N_c\alpha_{\rm s}}{4T} \right) \frac{1}{r} \right], \tag{3.10}$$

$$\overline{\Gamma}^{\uparrow}_{o\to o} = \hat{\kappa}T^3 \frac{N_c^2-4}{2(N_c^2-1)} \frac{l+1}{2l+1} \left[r^2 - \frac{2l+3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} \right], \tag{3.11}$$

$$\overline{\Gamma}^{\downarrow}_{o\to o} = \hat{\kappa}T^3 \frac{N_c^2-4}{2(N_c^2-1)} \frac{l}{2l+1} \left[r^2 + \frac{2l-1}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} \right], \tag{3.12}$$

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with

$$\overline{\mathcal{D}}^2 = -\frac{\partial^2}{\partial r^2} + \frac{l(l+1)}{r^2}\,. \tag{3.13}$$

Steps 5 and 6 in section 3.4 should be replaced by

5. Initiate a *quantum jump* by generating an additional random number p_2 uniformly distributed in $[0, 1]$. This number will be used to select the jump operator to apply.

6. If the color state is singlet ($c = 0$):
   (a) Set the color state to octet, i.e. $c \to 1$.
   (b) Calculate the jump probabilities P_1 and P_2 from eqs. (3.7) and (3.8) as $P_{1(2)} = \langle\overline{\Gamma}^{\uparrow(\downarrow)}_{s\to o}\rangle / (\langle\overline{\Gamma}^{\uparrow}_{s\to o}\rangle + \langle\overline{\Gamma}^{\downarrow}_{s\to o}\rangle)$.
   (c) If $p_2 < P_1$, set $l \to l+1$, else $l \to l-1$.

   Else ($c = 1$):
   (a) Evaluate the transition amplitudes $\overline{\Gamma}_1 = \langle\overline{\Gamma}^{\downarrow}_{o\to o}\rangle$, $\overline{\Gamma}_2 = \langle\overline{\Gamma}^{\uparrow}_{o\to o}\rangle$, $\overline{\Gamma}_3 = \langle\overline{\Gamma}^{\downarrow}_{o\to s}\rangle$, and $\overline{\Gamma}_4 = \langle\overline{\Gamma}^{\uparrow}_{o\to s}\rangle$ using the operators listed in eqs. (3.9)–(3.12) above. From these, obtain the jump probabilities $P_i = \overline{\Gamma}_i / \sum_{j=1}^{4} \overline{\Gamma}_j$.
   (b) If $p_2 < P_1$: $l \to l-1$,
   else if $P_1 < p_2$ and $p_2 < P_1 + P_2$: $l \to l+1$,
   else if $P_1 + P_2 < p_2$ and $p_2 < P_1 + P_2 + P_3$: $l \to l-1$ and $c \to 0$,
   else if $P_1 + P_2 + P_3 < p_2$: $l \to l+1$ and $c \to 0$.

We note that the phenomenological results presented in figures 4 and 5 of the manuscript are unaffected as the partial widths $\Gamma_{s\to o}$, $\Gamma_{o\to s}$, and $\Gamma_{o\to o}$ used to construct the effective Hamiltonian are unchanged.

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